What a Single-Horizon Comparison Cannot See: Exposure Concentration in a Learned-Ranker Matching Market, and How Long It Lasts

Abstract

A recommender that decides which candidates a matching market's agents ever see changes what a preference list means. This paper shows that measuring the harm at one horizon can invert the conclusion. In a simulated market of 1000 agents whose top-20 shortlists come from a neural collaborative filter, exposure Gini under the learned ranker exceeds an oracle's by $+0.0237$ $[+0.0199, +0.0274]$ at a hundred epochs over twenty seeds, and is positive on all twenty of a fresh seed set. Extending to three hundred shows no floor: the gap peaks at $+0.3055$, and a pre-declared equivalence test at a 0.002 margin puts the tail inside it, so the hundred-epoch number describes a transient rather than a bias, though a flat tail is also what a very slow decay looks like. The ordering survives all 12 sweep cells but varies fivefold in size, and the mechanism usually offered for the decay is not established here: the two interventions meant to test it do not. A companion institutional optimum exists only in a configuration that does not respond to the market. Five seeds carry the horizon, sweep and institutional results and twenty carry the shipped-horizon ones; nothing is calibrated.

1 Introduction

When a recommender sits between the two sides of a matching market, it decides which candidates anyone ever sees. The consequence is easy to state and awkward to price: a preference expressed over a truncated list is not the same preference that operates over the whole population. The agent on nobody's list is not rejected; she is never considered, and no preference data will record the difference.

The measurement problem

The standard way to measure that harm is to compare exposure under the learned ranker with exposure under a ranker that is perfect at the same task, on the same population and seed, and to report the difference at some horizon. This paper shows that measurement can invert its own conclusion. In the market simulated here the gap between the learned ranker and an oracle is large and positive at twenty epochs, small and still positive at a hundred, and indistinguishable from zero at three hundred. A study that stopped at either of the first two horizons would report a standing deficiency of learned rankers where what exists is a start-up cost, and the two are not the same finding: a standing bias is a reason to regulate the ranker, a transient is a reason to wait.

The inversion is not an artifact of one number being misread. It is what a single-horizon comparison cannot see, because a decay and a floor look identical from inside either one. The distinction is measurable, since it takes a longer run and this paper runs one, and establishing that it changes the answer is the paper's central claim.

The result

A finite two-sided market of a thousand agents carries a four-dimensional trait vector and a private commitment type. A neural collaborative filter trained on the swipe log produces a top-20 shortlist per agent, and a truncated deferred-acceptance pass clears the market over those shortlists and nothing else.

The gap is a transient. The learned ranker's exposure Gini exceeds an oracle's by +0.0237 [+0.0199, +0.0274] at the shipped hundred-epoch horizon over twenty seeds, positive on all twenty and on all twenty of a fresh set. Extended to three hundred epochs it peaks at +0.3055 shortly after the start, is +0.02240 at epoch 99, and is below 0.002 from about epoch 198 to 262 depending on seed; a pre-declared equivalence test at that margin puts the tail inside it. What is not established is the limit, since a flat tail is also what a very slow decay looks like (§5.3, §5.4).

It is real everywhere but not constant in size. The gap is positive with an interval excluding zero in all 12 sweep cells, on both interval constructions, and reverses nowhere; its magnitude varies by a factor of 5, so no single number is "the" gap (§5.6).

The explanation is a hypothesis this study does not establish. The mechanism offered is that the filter learns whom it can represent. That is consistent with the trajectory and untested here: the cell meant to isolate a market without coverage learning turns out to be the same process running more slowly (reach 0.49 by epoch 299 against 0.46 at epoch 99), and freezing the ranker freezes its shortlists, so any frozen arm is constant by construction (`docs/PREREGISTRATION.md` E3).

The institutional companion does not survive where the paper runs. An interior welfare optimum exists in the gate regime that ignores the market's own outcomes, and replicates its location on fresh seeds; re-run at the shipped gate, no cell of the forty beats the global market at any size or cost (§5.5, Appendix A).

What is new

1. **A horizon dependence that inverts a conclusion.** A single-horizon comparison of a recommender's exposure effect cannot distinguish a transient from a standing bias, and in this market the distinction reverses the interpretation. §5.3, §5.4.

2. **A separation of what replicates from what does not.** Twenty fresh seeds that took no part in any earlier choice reproduce every ordering, and reproduce the institutional optimum's location while cutting its claimed precision by a factor of about five. §5.2, Appendix A.

3. **A regime finding that refutes the institutional result where the paper actually runs.** The interior optimum is a property of a gate configuration that does not respond to the market; at the shipped gate no cell of forty beats the global market. §5.5.

4. **A mechanism claim reported as untested.** The interventions that were meant to test it do not, and the paper says which and why. §5.6.

Scope, the closed-form context, and what this is not

Six hypotheses were posed in advance, each with a stated kill condition, and five kinds of test applied: seed-bootstrapped intervals, a horizon extended to convergence, a twelve-cell parameter sweep with its thresholds fixed before the cells were read, multiplicity correction over the forty institutional cells the headline optimum was selected from, and a replication on seeds chosen for the purpose rather than inherited. §4 states the sample sizes so every number can be checked.

Nothing here is a claim about a population. Every parameter is a modeling choice, nothing is calibrated, twenty seeds carry the shipped-horizon quantities and five carry the horizon, sweep and institutional results, and one population size carries it. The results bound what this mechanism can do under a stated utility; §6 says so at length rather than in a footnote.

There is also a specific absence the paper is built around. The constraint here is *visibility*, not compatibility: the feasible set is the top-k of an estimated multi-attribute utility, lists differ by agent, and estimator error is correlated in a way no marginal captures. For a single scalar notion of status the aggregate questions do have closed-form answers, and a recent preprint argues they are settled [Bilderbeek, 2026]: with unions constrained to be hypergamous, the minimal unmatched mass is a functional of the two marginal status distributions alone, and no behavioral strategy changes it. That result depends on the constraint being a nested upper set, which is what makes the deficit a one-dimensional supremum, and a learned top-k list has neither property. The structure that makes the closed form tractable is absent here rather than merely harder, which is why the question has to be answered numerically, and why the answer turns out to depend on *when* it is asked.

2 Related work

2.1 The closed-form result, and what it settles

For a marriage market where status is a single scalar and unions must be hypergamous, the aggregate questions have closed-form answers. A recent preprint argues the picture [Bilderbeek, 2026]: universal hypergamy is feasible exactly when one side's status distribution first-order stochastically dominates the other's, and when it is not, the minimal unmatched mass is the hypergamy deficit $D = \sup_t (G_W(t) - G_M(t))_+$. The finite-population condition underneath it is Hall's marriage theorem [Hall, 1935], of which this deficit is the maximum deficiency; the preprint's contribution is the continuum form and its application. This is checked against the simulator below rather than taken on trust. No behavioral strategy changes either fact, status gains are transfers of rank rather than creations of it, and under status-ranked preferences the stable matching is unique and positive assortative. Existence of a stable matching is Gale and Shapley [1962]; uniqueness under a common ranking is a property of the lattice they generate, set out by Knuth [1996], because when every agent on a side ranks the other side identically the proposer-optimal and acceptor-optimal matchings coincide.

The consequences for anyone modeling this are severe. Unmatched mass is a functional of the two marginal status distributions alone, so interventions that rearrange who meets whom without changing those marginals cannot move it. This paper does not re-derive the result; it reproduces it against the simulator (§4.5) and then changes the constraint.

2.2 The empirical record, and why a model is the instrument

The empirical literature is consistent and mostly about marginals. Becker [1973] supplies the complementarity condition: supermodular surplus implies positive assortative matching, so measured sorting is evidence about the surplus function. Schwartz and Mare [2005] trace US educational homogamy from 1940 to 2003, falling then rising. Eika et al. [2019] find assortative mating accounts for a non-negligible share of cross-sectional household income inequality while *changes* in assortative mating barely move trends: a level effect without a trend effect, and a warning about which one a cross-sectional model can speak to. Esteve et al. [2016] show the educational gender-gap reversal shifting marriage patterns toward hypogamy with no preference change required.

On the micro side, Fisman et al. [2006] randomise speed-dating encounters and find trait weights differing by sex and selectivity rising with group size; Lin and Lundquist [2013] show trait salience is not separable, which bounds the additive utility used here. In online settings, Hitsch et al. [2010] estimate mate preferences, feed them into deferred acceptance, and find predicted stable matches track observed ones closely, concluding that preferences rather than frictions explain most sorting. Bruch and Newman [2018] find a desirability hierarchy consistent across cities, with reply probability falling sharply in the desirability gap.

Why model rather than estimate? Because every instrument above observes *realized* matches inside one institution, and none can answer a counterfactual about a different one. The question here is institutional: what happens when the list an agent sees is produced by an engagement-optimizing estimator rather than given. Kanoria and Saban [2021] and Halaburda et al. [2018] take the same view from the design side, deriving platform welfare from a model of restricted choice rather than estimating it. Galichon and Salanié [2022] make the structural position precise for identification, which is why this paper reports welfare only inside the model. The precedent for computing when the closed form does not apply is the clearinghouse work of Roth and Peranson [1999].

2.3 Rankers, feedback loops, and the visibility constraint

He et al. [2017] establish the collaborative-filtering architecture used here and make no claim about markets. Pizzato et al. [2010] introduce the reciprocal recommender and show that accounting for both sides lifts top-ten success substantially; Tomita et al. [2022] describe deploying a matching-theory-based recommender on a live platform. Reciprocity-aware ranking is solved engineering, and a paper claiming novelty for its ranker would be claiming novelty for something built in 2010. Chen et al. [2023] go further, running a matching-model-based ranker on a live platform and measuring realized matches. That is the strongest constraint on any novelty claim here: same intersection, different question.

The loop literature is more directly adversarial. Chaney et al. [2018] show that training on data confounded by the system's own recommendations homogenises users without raising utility. Mansoury et al. [2020] show a loop amplifying popularity bias and hitting minority-group users hardest. Jiang et al. [2019] separate the echo chamber from the filter bubble. Abdollahpouri [2019] frames popularity bias as a fairness problem, and Sonboli et al. [2022] show that in multisided settings fairness criteria for different stakeholders conflict.

Every one of these predicts a loop that *persists or compounds*. The result here points the other way, and the distinction this paper offers is between a ranker's deficiency being coverage, which training could fix, and preference distortion, which it would not. That distinction is not in the loop literature, and the honest state of it after this study is that it remains a hypothesis: the trajectory is consistent with it, and neither of the interventions run here separates training from reaching (§5.6, `docs/PREREGISTRATION.md` E3). A reader should check it hardest of anything in the paper, and the paper reports that it does not survive being checked.

2.4 Cold start, exploration, and the dynamic market

Three literatures bound the question and are not engaged elsewhere in this section. The first is *cold start* in recommender systems: Schein et al. [2002] define the problem as one where recommendations are required for items no user has rated, and the standard taxonomy separates item from user cold start. The gap measured here is a cold-start quantity in exactly that sense, because the ranker is being asked to rank agents it has no interactions for, but the literature's usual remedies

(content features, hybrid models, side information) are not what the trajectory shows happening, which is the ranker accumulating interactions until coverage saturates.

The second is *exploration*. Li et al. [2010] treat the choice of what to show as a contextual bandit and make optimism about unseen items the mechanism by which an estimator learns. That is precisely the intervention this paper could not implement: an explore-then-exploit ranker would deliberately show low-estimate agents, which would move reach directly rather than through training. None of the four arms here explores. A reader who expects the bandit framing should know that the shipped ranker is pure exploitation, that this is why the cold start exists at all, and that the paper does not test whether exploring would remove it.

The third is *dynamic matching markets*. Akbarpour et al. [2020] model a market where agents arrive and depart stochastically and study the trade-off between thickness, information and waiting. This paper's market has neither arrivals nor departures: generational turnover replaces agents but nobody enters an otherwise empty market and nobody leaves in response to an outcome. That is why every result here is about a transient that ends, rather than about a steady state that could persist. In a market with continuous arrivals some share of the population is always at epoch zero, and the steady-state gap is the object of interest there. It has not been measured (`docs/ROADMAP.md` §1).

Two further results bound what mediation can be credited with. Roth and Rothblum [1999] show truncating a preference list changes outcomes strategically, so a top-k list is not a neutral convenience. Ashlagi et al. [2017] show that in an unbalanced random market even a slight imbalance produces an essentially unique stable matching with the long side matched near-randomly, so some measured asymmetry is competition rather than curation. Both cautions apply: the truncation here is imposed rather than chosen, and the market is balanced, so the competition channel is held at zero by design rather than measured away.

2.5 Institutional design and the selection problem

Kanoria and Saban [2021] show a platform that restricts agents' actions can raise welfare. Halaburda et al. [2018] show a platform can compete by limiting the choices it offers, because restricting one agent's choice also restricts her competitors'. Hoppe et al. [2009] give the sharpest constraint on the interior optimum sought here: with two-sided incomplete information the output gain from assortative matching can be exactly offset by signalling costs, which is what failure at high signalling cost looks like.

That literature also names the objection this paper had to design around. If an institution excludes agents, its measured welfare improves for the arithmetic reason that excluded agents are no longer counted. The comparability rule of §3 is therefore a response to a known objection, not a safeguard invented here, and both accountings are reported so the artifact is visible.

Two results bound any W-side number. Roth [1982] establishes that no procedure is both always stable and always strategy-proof, with the proposer-optimal matching simultaneously the acceptor-pessimal one; Knuth [1996] supplies the lattice structure that makes those one statement. Since hypergamy lives in the acceptor's utility, this bias lands where the paper's question is, and every acceptor-side level reported is a lower bound across stable matchings.

2.6 What is new here

Thirty-eight sources were examined for whether any already does recommender-mediated matching on multiple traits. Thirty-five do not. Three require the claim restated: Chen et al. [2023], Pizzato et al. [2010] and Tomita et al. [2022] between them rule out any novelty claim resting on the ranker.

The gap that survives is narrower than "recommender-mediated matching" and is a conjunction of three specific absences: no characterisation of unmatched mass against the visibility constraint, no institutional comparison on a common welfare basis, and no horizon. What this paper adds is the recommender-intermediated setting in which visibility is a truncated top-k list over an estimated multi-attribute utility and no closed form is known. It claims nothing for its ranker, and no progress on the multidimensional-status, search-friction or endogenous-status problems the closed-form literature names as open.

A note on verification

Reproducing a closed form is verification: it establishes that the machinery is correct, not that the model describes anything. Windrum et al. [2007] and Fagiolo et al. [2007] set out why empirical validation of an agent-based model is a family of practices rather than a single test, and why calibration alone does not resolve it. This paper claims verification and declines the validation question.

3 Model

The model is a finite two-sided market cleared once per epoch, with a learned ranker standing between the two sides. Three layers run in a loop: agents form preferences, a collaborative filter decides what each agent can see, a matching algorithm clears the market over what was visible, and the clearing outcome feeds back into the acceptance threshold that gates the next epoch's visibility. This section describes each layer, the two institutional environments the paper compares, the utility that drives them, and the condition under which the feedback channel is live rather than decorative.

3.1 The agent layer

Agents are split into a proposer side M and an acceptor side W of equal size. Each carries a four-dimensional state vector $[s, a, c, t]$: status, attractiveness, sociability and age. Status is drawn lognormal; the other three are clipped to the unit interval. Every trait is z-scored against population statistics before it reaches a utility, so each weight below is expressed against a unit-variance trait and the weights are directly comparable to one another. The choice to z-score is load-bearing for interpretation: it means only the *rank order* of a trait within the population matters, and a distribution with no variance would annihilate its term.

Two further agent attributes drive the institutional comparison. A private *intent* type $\tau_i \in (0, 1]$ is each agent's willingness to commit, invisible to others. A pairwise *alignment* $g_{ij} \in [0, 1]$ measures value compatibility and enters the dissolution hazard rather than the utility.

3.2 The recommender layer

A neural collaborative filter with a generalized matrix factorisation path and an MLP path is trained on the swipe log, the record of which pairs were liked, and scores every unseen pair. Each agent then keeps a top-k shortlist of length $k = 20$ from those scores, and *nothing outside that shortlist is visible to the matching step.* This is the architectural commitment the paper rests on: the filter is not a convenience wrapper around a preference list, it *is* the preference list. It is also why a preference expressed over this list is not the preference the same agent holds over the population, which is the point of departure for the whole exercise.

The ranker is deliberately off-the-shelf. It is cited for the estimator and not for a result about markets, and the contribution would be void if it depended on a novel architecture.

3.3 The matching layer

Clearing is a truncated deferred-acceptance pass over the shortlists, returning the matched pairs and a blocking-pair count. Truncation matters: with lists truncated at k, the procedure is the classical algorithm run on top-k lists rather than on full lists [Gale and Shapley, 1962], and the classical stability guarantee does not carry over unchanged [Roth and Rothblum, 1999]. It also means the clearing bias is a property of the procedure class and not of the algorithm specifically, as the wage-forming counterpart of the same procedure shows [Kelso and Crawford, 1982]. A full-information arm ranks by true utility under the same proposer direction and the same shortlist length, and is the benchmark against which algorithmic loss is measured.

Two properties of the clearing step bound what any W-side level can mean. Deferred acceptance with M proposing returns the proposer-optimal matching, which is simultaneously the acceptor-pessimal one: among all stable matchings, every acceptor receives her worst [Roth, 1982, Knuth, 1996]. Since hypergamy lives in the acceptor's utility, this bias lands on exactly the side the paper is about. The full-information and mediated arms use the same direction, so the *gap* between them is far less affected than either absolute level. Any W-side level reported here is a lower bound across stable matchings and must never be compared against a W-proposing baseline without saying so.

3.4 The utility

Utility is additive over four terms: a status term, an attractiveness term, a distance penalty, and an indicator bonus for a status surplus. The bonus is what makes the model hypergamic rather than merely assortative. It rewards a partner at or above the agent's own status by a margin δ, and is the model's operationalisation of the requirement that a partner rank at or above you rather than beside you.

Two parameters modify the utility's realized shape. A per-agent lognormal multiplier with coefficient of variation σ_{taste} introduces taste heterogeneity, so different agents weight the same traits differently, which is the same device by which idiosyncratic tastes are added to a transferable-utility marriage model [Choo and Siow, 2006]. A persistent pairwise chemistry shock with standard deviation σ_{shock}, drawn once per ordered pair and held across epochs, enters in absolute utility units. Neither has an empirical anchor, and §5.6 treats both as knobs whose value changes the headline rather than as estimates.

3.5 The two environments

Environment A is the global market: a large population, visibility mediated globally by the learned ranker, no signalling cost, no entry filter.

Environment B is a subgraph of the *same* population. Agents are drawn into a pool of size at most ϕN, filtered on an observable value score, and within the pool visibility is complete, since every entrant sees every other, so B has information and no liquidity where A has liquidity and no information. The comparison is therefore a comparative static in an *institutional* parameter at a fixed population, which is a different exercise from the comparative statics in market size that the matching literature already treats [Crawford, 1991]; changing the subgraph size does move one side's size, and that channel is separated from the institution effect by holding the population fixed and varying only the shares. Proposing in B costs $c_B > 0$ and succeeds with probability p,

so a proposal's presence carries information about the proposer's intent. A defection penalty d is charged to a defector after a match dissolves.

The signalling mechanism is the model's core. An agent proposes only when $\tau_i V_{ij} p > c$, so the set of proposers is exactly $\{i : \tau_i > c/(V_{ij}p)\}$, and the separating threshold is

$$\tau^*(c) = \frac{c}{\bar{V}p}. \tag{1}$$

At $c \to 0$ the threshold vanishes, every type proposes, and a received proposal carries no information about intent: the posterior equals the population mean. This is a pooling outcome reached through a *cost* channel rather than through unobservable quality [Akerlof, 1970], and the paper treats it as an analogy to the classical lemons result rather than as an application of it. At $c > 0$ the threshold is positive, low-intent agents exit by their own incentive to avoid the cost, and the posterior rises: a separating outcome in the single-crossing sense [Spence, 1973]. A and B are therefore two settings of one parameter, not two theories.

An agent who cannot infer intent from a proposal must use something else, and the model's rational response is to raise the bar on observables. That is the acceptance threshold θ, and its derivation from the pooling result rather than its assumption as an exogenous contagion is the content of H2.

3.6 Welfare, and the comparability rule

Welfare is defined per epoch over the whole population: the value created by matched pairs, less the signalling cost spent on proposals, less the loss from dissolved matches including the defection penalty.

The accounting rule is what makes the comparison meaningful. Agents excluded by B's entry filter are not deleted, and agents who fail to match inside B are not dropped: every agent appears in the welfare sum exactly once, and an agent B excludes is scored at whatever A would have given them. Without this rule an entry filter would look like a gain for the purely arithmetic reason that it removes the agents it cannot serve. This is a response to a known objection rather than a safeguard the project devised, since restricting choice can raise the welfare of those inside the restriction [Halaburda et al., 2018], and the model prices the cost to non-participants instead of ignoring it.

Both accountings are computed. The common one, over all N agents, is the number the paper reports. The selected one, over entrants only, is computed so that the selection artifact can be reported honestly rather than hidden, and is never quoted as a headline.

3.7 The θ gate and its binding condition

The one feedback channel in the model runs from clearing outcomes to visibility. After each epoch, an acceptor whose received proposal count is informative updates θ; θ is the quantile of her own utility distribution above which a candidate is admissible, so raising θ shortens her admissible set and lowering it lengthens it.

The gate binds only under a condition that is worth stating exactly, because several results below are regime-specific and the regime is set by it. Let ϕ be the target admissible fraction, the share of candidates an agent finds admissible at epoch 0, and let k be the shortlist length. An agent's admissible set at epoch 0 has about ϕN members. The gate can only truncate an agent's visibility if that set is *shorter* than the shortlist, since otherwise she takes the top k of a larger pool

wherever θ sits and the update moves nothing. So the gate binds when

$$\phi < \frac{2k}{N}. \tag{2}$$

At the shipped values $k = 20$ and $N = 1000$ this is $\phi < 0.04$. The config ships $\phi = 0.02$, inside the binding region; the earlier shipped value was $\phi = 0.2$, outside it, and the ablation data for that regime is labelled *inert* throughout. The factor of two in (2) reflects that the relevant comparison is against the candidate set on each agent's own side of the market rather than against the whole population.

Three consequences follow, and they are used later. First, ϕ and k interact: raising the shortlist length or lowering the population requires lowering ϕ to keep the gate live, and the configuration validator warns when the rule is broken. Second, the difference between an inert run and a live run is a statement about whether the market responds to its own clearing outcomes, so an ablation measured at $\phi = 0.2$ is not a noisy version of one measured at 0.02; it is a different dynamical system. Third, the gate's liveness is a property of one epoch's calibration, and it relaxes: the update pulls θ toward a yield in the unit interval from an initial value far above it, so the gate truncates at the start and stops truncating within a few epochs. That relaxation is why the live and inert regimes produce nearly identical final-epoch gaps despite different trajectories.

Those three facts decide one of this paper's results rather than merely qualifying it. The institutional comparison was measured at $\phi = 0.2$, outside the binding region, and re-run at the shipped $\phi = 0.02$ it reverses (§5.5). A reader who wants to know why a small arithmetic condition in this section matters at the level of a conclusion has it here: an inert gate means the market is deaf to its own outcomes, and an institution measured in a deaf market is not being compared under the same learning dynamics as one measured in a live one.

4 Design

This section states what was run, at what sample size, against what controls, and what was held fixed, so a reader can tell which numbers in §5 are comparable to which and can re-derive any of them from the shipped configuration without re-running the simulation.

4.1 Regimes, and why the horizon is a design variable

The shipped configuration is $N = 1000$ agents, shortlist length $k = 20$, gate target $\phi = 0.02$, nominal horizon 100 epochs. Five regimes appear, and the differences between them are results in their own right:

regime	ϕ	epochs	what it isolates
`inert20`	0.2	20	gate never binds; market deaf to its own outcomes
`live20`	0.02	20	gate binds, then relaxes
`live100`	0.02	100	the shipped horizon
`base20`	0.02	100	the same horizon, twenty seeds
`long300`	0.02	300	convergence, not a steady state

The institutional comparison uses a sixth: $\phi = 0.2$, the inert gate, at five epochs per paired run. That is the same gate as `inert20` and it is the regime in which H4 holds and in which it is refuted at the shipped gate (§5.5).

The inert regime is retained rather than discarded because the difference between it and the live regime is the evidence that the feedback channel matters at all. The two 20-epoch regimes differ in

ϕ alone; the two live regimes differ in horizon alone, and their first epochs agree only approximately. The four arms of the 300-epoch run and the 100-epoch run are bit-identical through epoch 4 and then diverge in the NCF arm alone, reaching 3×10^{-4} in exposure Gini by epoch 19; the other three arms stay bit-identical. That is the signature of a different floating-point reduction order in the training path, not of a different model, and it is why every figure in this paper names the run it came from: `ablation_h100` and the committed `rigor/long_horizon` are two runs and not one run at two lengths. Their seed-42 NCF Ginis at epoch 99 differ in the third decimal (0.6935 against 0.6967), which changes no verdict, but the second directory cannot be regenerated from this code at any thread count tested (`docs/REVIEW_RESPONSE.md`, A3). The 300-epoch trajectory this paper quotes is therefore `rigor/long_horizon_1t`, the same design regenerated at one thread and checked against the committed run before it was quoted (`docs/PREREGISTRATION.md` E7).

4.2 Arms

The A-side ablation compares four arms on a shared population, seed and utility set, so differences are attributable to the visibility rule rather than to sampling:

- **ncf**, the learned ranker. The treatment arm.
- **oracle**, which ranks by true utility under the same proposer direction and shortlist length. The benchmark, not a deployable institution.
- **popularity**, which ranks by aggregate engagement; its exposure Gini sits at the analytic closed form, which checks the exposure machinery.
- **random**, which ranks uniformly, because distance from uniformity is not a measure of harm and without it a reader cannot tell whether the ranker's concentration is unusual or what any non-personalised rule does.

The institutional comparison adds Environment B at four signalling costs ($c_B \in \{0.01, 0.05, 0.10, 0.25\}$) crossed with ten subgraph sizes ($\phi_B \in \{0.02, 0.05, 0.10, 0.15, 0.25, 0.35, 0.50, 0.65, 0.80, 1.00\}$): forty cells, five seeds, both accountings. Environment A there is the *oracle* arm, so nothing is estimated in either environment; the NCF-mediated comparison is reported separately and reverses the ordering. The drift decomposition is a 2×2 factorial over exogenous drift and algorithmic mediation, each on or off, from which the three terms are estimated.

4.3 Sampling and what counts as a replicate

Five seeds (42 through 46) underlie the horizon, sweep and institutional results, and twenty the shipped-horizon ones. This is the design decision most likely to be objected to, so the response is registered rather than deferred. A seed is a population, not a run: each draws fresh traits, intents, alignments and pairwise shocks and holds them fixed across arms, so arms are compared *paired* by seed and every gap is a per-seed difference.

Which quantities rest on twenty seeds and which on five. So that a reader is never guessing: the exposure Gini, the NCF–oracle gap, the clearing rate, rank satisfaction, reach and algorithmic loss at the shipped hundred-epoch regime are all quoted from a twenty-seed run. Everything else is five. That includes every horizon claim (the 300-epoch series, its decay fits, its asymptote and its stationarity tests), every sensitivity-sweep claim (twelve cells, five seeds each), the institutional grid and its optimum, and the drift decomposition. A number in §5 that is not

one of the six named above is a five-seed number, and the text says so where the distinction could mislead.

A twenty-seed run (42 through 61) was executed at the shipped regime. A power calculation against the twenty-seed standard deviation requires fewer than five seeds to detect every headline effect at 80% power, with raw requirements below one in every case, and no headline changes sign or crosses zero at any seed count from five to twenty (`docs/STATS.md` §2.4). That is a statement about *sign*, not *width*. Two quantities behave badly on width: the five seeds are the minimum-variance contiguous block of the sixteen available for algorithmic loss and second-lowest for clearing, so their standard deviations understate the population's by roughly a factor of three and two and a half (§2.2). Every interval on those two is reported from the twenty-seed run, and no headline rests on a five-seed interval for either.

4.4 Intervals, multiplicity and the sweep

The estimating method is fixed before results are read and differs by seed count for a measured reason. At five seeds the Student-t interval is quoted; at twenty, the BCa bootstrap with 10,000 resamples. The BCa interval is reported beside the t interval at five but never quoted alone, because at that size its half-width is narrower than the sample's own standard deviation (`docs/STATS.md` §1). Bootstraps are seeded from a fixed constant and regenerate byte for byte.

Two families are corrected. The institutional grid holds 200 rows but 40 distinct cells (ten sizes by four costs), each replicated over five seeds; a cell is what was selected over, so its seeds are replicates rather than five further chances to win, and the family is 40. Uncorrected family-wise error under independence would be 0.871, but the cells are not independent, since adjacent sizes share populations and the same seeds are reused, so the realized rate is lower and is not claimed; the nominal figure is quoted as an upper bound. The decomposition corrects over its three terms.

Every committed result sits at one setting of four knobs, each swept at three levels with the others shipped: taste heterogeneity $\sigma_{\text{taste}} \in \{0.2, 0.5, 1.0\}$, chemistry noise $\sigma_{\text{shock}} \in \{0.5, 1.5, 3.0\}$, the weight ratio $\beta/\alpha \in \{1, 2, 3\}$ and the hypergamy margin $\delta \in \{0, 0.5, 1\}$: twelve cells, five seeds, 100 epochs. Four claims were fixed with their thresholds before the cells were read, so a reader can disagree with the thresholds rather than only with the verdicts: that the gap excludes zero and is positive (C1), that its point estimate is at most 0.05 (C2), that the ranker reaches at least 75% of agents (C3), and that clearing is non-degenerate (C4). C2's ceiling and C3's floor are the two judgments most exposed to the charge of being chosen after the fact, and the only mitigation is that they were written down first. The shipped cell appears in three axes with identical numbers (+0.0260, reach 0.9566), which checks that the axes are independent.

4.5 Validation against closed-form results

Before the recommender is asked anything, the matching and survival-function machinery is checked against proved results. This is verification, not validation: agreement with a theorem shows the code is correct and says nothing about whether the model describes a population.

Two tools are used because the results are of two kinds. The deficit results concern the largest matching under a *compatibility* constraint, which is a maximum-cardinality bipartite matching problem and is solved as one; a stable-matching procedure does not compute it. Its finite form is the deficiency version of Hall's marriage theorem [Hall, 1935], of which the preprint's continuum deficit is the limit. The uniqueness result is the stability question and is tested with deferred acceptance [Gale and Shapley, 1962]; that a common ranking makes the stable matching unique is the collapse of the stable-matching lattice to a point [Knuth, 1996].

Check	N	Measured	Evidence	Verdict
Thm 2.2, deficit D	$N = 300$	exact on 5 seeds	matching = empirical sup	agrees
Example 2.4	$N = 20{,}000$	0.2519 vs 1/4	single seed, tol 0.02	agrees
Thm 3.3, $D(14.0)$	$N = 5{,}000$	0	exact	agrees by construction
Thm 3.3, $D(30.48)$	$N = 5{,}000$	0.7254	formula 0.7217	agrees
Thm 3.3, unequal var.	$N = 5{,}000$	0.7433	formula 0.7380	agrees
Thm 5.1, unique	$N = 300$	same matching both ways	0 blocking pairs, 5 seeds	agrees
Thm 5.1, perturbed	$N = 300$	22 / 8 / 11 / 12 / 8 pairs differ	uniqueness fails, stability holds	as the theorem requires

Table 1: Validation against the closed-form results: the deficit identity and height-deficit form in the first five rows [Hall, 1935, Bilderbeek, 2026], assortative uniqueness and its precondition in the last two [Gale and Shapley, 1962, Knuth, 1996]. The theorem numbers are the preprint's; the deficit and uniqueness results themselves are classical. Generated by `paper/tables.py` from `data/processed/rigor/validation.csv`, which `scripts/run_validation.py` writes by recomputing the quantities `tests/python/test_validation.py` asserts; a test compares the committed file against a fresh computation, so the table cannot silently disagree with the checks it reports.

Table 1 reports both. The most informative check is the last: adding noise of order one percent of the status scale to each agent's own view of the other side leaves stability intact, with zero blocking pairs in both proposer directions, while uniqueness fails. The theorem's conclusion therefore rests on the lists being *common*, not merely correlated with status.

What this licenses is narrow: it shows the deficit and stability machinery is correct, and does not show the recommender's unmatched mass has a closed form. The reasons are structural. The deficit is a supremum because the compatibility constraint is a nested upper set on a single scalar, and a top-k list from an estimated multi-attribute utility has neither property.

4.6 What was not run

Three gaps are stated rather than left for discovery. The 300-epoch convergence run exists only at the shipped cell, so convergence is established at one point and not across the sweep, which is 100 epochs deep everywhere. The full-information benchmark materialises an $O(N^2)$ preference matrix and is capped at $N = 5000$, so larger-population results are NCF-only and lose their reference. And the model has no attrition mechanism, so one hypothesis cannot be tested at all (§A.5).

5 Results

The results are reported by hypothesis, and the split is deliberate. Two carry the paper's argument and are given in full here: H3's ordering and the cold-start result that qualifies it. Five are summarized in one paragraph each, with the full analysis in Appendix A, because their verdicts are weaker and a reader should be able to weigh the central claim without them. The convergence result gets its own subsection because it asks about a limit rather than testing a hypothesis. Table 2 collects the four hypotheses with a committed interval; H1, H2 and H6 have none, for reasons stated in place.

Two conventions hold. Every interval is a 95% interval over the seed distribution, and table 2 prints both the Student-t and the BCa bounds for every row rather than choosing one silently, because which is preferred is a small-sample question and the two disagree at five seeds. The t interval is the one with nominal coverage at n = 5; the BCa bootstrap attains roughly 84% there and returns a half-width narrower than the sample's own standard deviation (`docs/STATS.md` §1). At twenty seeds the two agree to 0.002. Where a row's interval has been widened for multiplicity,

Hyp.	Regime	Measured	t interval	BCa interval	Verdict
H3	NCF − oracle Gini (base20, 20 seeds)	+0.0237	[+0.0199, +0.0274]	[+0.0204, +0.0271]	survives; the 100-epoch headline
H3	NCF − oracle Gini (live100, 5 seeds)	+0.0251	[+0.0155, +0.0347]	[+0.0186, +0.0307]	survives; size is horizon-dependent
H3	same gap (long300, 5 seeds)	+0.00044	[-0.00068, +0.00155]	[-0.00029, +0.00105]	below 0.001 from about epoch 250
H4	welfare B − A (c=0.01, sf=0.65; $\phi = 0.2$ inert gate)	+64.62	[+28.80, +100.43]	[+55.07, +78.46]	survives in this regime only; refuted at the shipped gate
H5	interaction term (factorial, 5 seeds)	+16.70	[-33.42, +66.82]	[-13.30, +53.05]	null under both; BCa includes zero too

Table 2: The hypothesis ledger: hypothesis, regime, point estimate, both 95% interval constructions, verdict. The t and BCa columns are printed side by side so that the choice between them is visible rather than implied; the t interval has nominal coverage at five seeds and is the one to read where they disagree. The H4 row alone carries a multiplicity correction (Bonferroni across 40 cells) and its verdict says so; the other rows are uncorrected and their families are stated in the text. That row's regime is marked because it matters: the institutional grid was measured at the inert gate ($\phi = 0.2$), and re-run at the shipped gate ($\phi = 0.02$) no cell of the forty beats the global market, and the cell below reads -17.23 there (§5.5). Every row traces to `data/processed/rigor/` and to §5.2, §A.3 or §A.4. H1, H2 and H6 are absent because no interval is registered for them; see §6.

which is the H4 row only, the caption and the row both say so: the H4 bounds are Bonferroni-t across the 40 institutional cells, and the uncorrected and Bonferroni-BCa bounds are beside them in the text (§A.3; `docs/STATS.md` §5.3). Gaps are always the paired per-seed difference, never a comparison of two arms' overlapping intervals.

5.1 H1, H2, H4, H5, H6: what was tested, and what happened

Five of the six hypotheses are reported in full in Appendix A, because their verdicts are weaker than the cold-start result and a reader should be able to weigh the argument without them. In one paragraph each:

H1 (a proposing cost selects on intent). Direction holds, with mean received intent rising monotonically in c, but the magnitude does not match the closed form, because the closed form evaluates its threshold at a prior success probability the market does not supply. Not refuted; no interval registered (§A.1).

H2 (threshold inflation as defence against pooling). Untestable as shipped: the θ update reads only a proposal count, never received intent, so no path exists by which the predicted slope could appear. Reported as an absence of mechanism, not as a refutation (§A.2).

H4 (an interior institutional optimum). Refuted at the configuration this paper runs. At $c_B = 0.01$ in the inert gate regime there is a non-monotone welfare difference with its optimum at subgraph size 0.65, which survives Bonferroni and Holm; re-run at the shipped gate, no cell of the forty beats the global market at any size or cost. The hypothesis therefore holds in a regime the rest of the paper does not use and fails in the one it does (§A.3; §5.5 below).

H5 (an interaction term). The drift-by-mediation interaction is +16.70, interval $[-33.42, +66.82]$, not distinguishable from zero, which is what H5 allowed. The algorithmic term dominates (§A.4).

H6 (differential attrition). Not tested at all: the model has no attrition margin, so no committed artifact exists and no number is quoted (§A.5).

5.2 H3: estimation error concentrates exposure

Under test. On a shared population and seed, the exposure Gini under the learned ranker exceeds that under an oracle ranking by true utility, because the ranker cannot represent every agent and those it cannot represent receive no exposure. Kill condition: the ranker matches or beats the oracle.

Measured. Not refuted. At the shipped regime and horizon, over twenty seeds, the paired gap is $+0.0237$ with a t interval of $[+0.0199, +0.0274]$, positive on all twenty (`docs/STATS.md` §1, row 2c; `intervals.csv`). The five-seed run gives $+0.0251$ $[+0.0155, +0.0347]$, positive on all five, which is the same quantity on a subset of the same seeds: seeds 42–46 are the first five of 42–61, so the two figures agreeing is expected and is not a test of anything. What it does establish is the interval's width, measured full width against full width: 0.0067 on the BCa construction at twenty seeds against 0.0122 at five, a factor of about 1.8, and that the point estimate moves by less than its own five-seed interval when the other fifteen seeds are added. The overlapping-sample agreement is not a test of anything; §5.6 is, and the ordering holds in all twelve of its cells on both interval constructions.

Refuted? No in direction; the substantive version, that estimation error is a *standing* source of concentration, is refuted by §5.3. The inequality holds; the mechanism attributed to it does not.

The replication on fresh seeds. Everything above rests on seeds 42–61, and `docs/STATS.md` §2.2 shows those five are an unusually low-variance block for two quantities. The gap itself is not one of them, but the objection is general and the answer is a fresh set. Twenty seeds (100–119) were registered as a confirmation set before they were run (`docs/PREREGISTRATION.md` E1) and executed in the same session, at the same thread count, as a regeneration of 42–61 for comparison. Every quantity listed below is a twenty-seed point estimate at 1 thread.

The fresh gap is $+0.0253$ $[+0.0217, +0.0290]$ against $+0.0225$ $[+0.0187, +0.0263]$ on the exploratory set: inside the pre-registered interval, and if anything slightly larger. The exposure Gini itself is 0.6722 against 0.6727, the oracle level 0.6468 against 0.6502, and the arm ordering (popularity $>$ ncf $>$ oracle $>$ random) holds on all twenty seeds. No quantity moved outside its interval and no ordering reversed, which is what E1 predicted.

Four measurements of this one quantity now exist, and a reader should know which is which. The headline, $+0.0237$, is the committed twenty-seed run in `rigor/base/`; it is quoted because a hundred-epoch claim should rest on the twenty-seed interval where one exists, and with the caveat that this directory no longer regenerates bit for bit (its seed-42 Gini differs from a fresh run at the same thread count in the fourth decimal). The five-seed $+0.0251$ comes from `ablation_h100/`, which does reproduce exactly, at 24 threads. The fresh $+0.0253$ and the regenerated exploratory $+0.0225$ come from this session at 1 thread. All four agree within each other's intervals (`docs/PREREGISTRATION.md` E8, E8a).

Two differences are worth naming because they bear on how much the seeds were worth. The fresh set has a smaller spread on the Gini quantities (sd 0.0111 against 0.0142 for the NCF level) and a larger one on algorithmic loss (0.0266 against 0.0222), which is the same quantity-specific behavior §2.2 describes rather than a contradiction of it. And the exploratory set re-derived here is not bit-identical to the committed `rigor/base/`: the seed-42 epoch-99 Gini differs in the fourth decimal. That directory predates the current code, so the comparison throughout this subsection is against the regenerated set, not against the committed one.

What it is not. The oracle ranks by true utility under the same direction and shortlist length: an upper bound on what this architecture could achieve, not a deployable alternative. A gap against an oracle measures how far a learned ranker falls short, not what a better-designed platform would do.

5.3 The cold-start result, and its convergence test

The question. H3's gap is positive at every horizon measured. Is it a floor or a transient? A hundred-epoch run cannot answer, because the gap is still falling at epoch 99. The run was extended to 300 epochs at the shipped configuration, five seeds (Figure 1; `data/processed/rigor/long_horizon_1t/`; `docs/STATS.md` §3).

Trajectory. The mean gap rises to a peak shortly after the start, then decays monotonically in the reported points:

epoch	0	6	19	49	99	124	199	250–299
gap	+0.2633	+0.3055	+0.2863	+0.1739	+0.02240	+0.00786	+0.00273	+0.000231

The peak is at epoch 6. The hundred-epoch figure is not a floor: at epoch 99 the gap is roughly a hundred times the residual it settles at, though both numbers are imprecise enough that the ratio is an order of magnitude, not a measured constant (`docs/STATS.md` §3.2).

Two cautions on reading the table. The points after epoch 124 are all within one final-epoch seed standard deviation of the NCF arm's Gini of each other (0.0108), so the decay is established to about epoch 124 and the later entries are ordered noise rather than a measured continuation: the claim that the series is *monotone* holds only of the points printed, not of the underlying trajectory. And the entries at 124, 199 and 250–299 are small differences of quantities whose own intervals are wide relative to their distance from zero, which is why no ratio is quoted as a measurement.

Convergence test. The tail mean over epochs 250–299 is +0.000231 with a 95% t interval of $[-0.000192, +0.000653]$, which *includes zero*: 3 of five seeds are positive ($t = 1.52$, $p = 0.204$; Wilcoxon $p = 0.312$). The tail is therefore indistinguishable from zero at five seeds, and no positive residual is claimed. What the data do support is the equivalence test in §5.4, which does not depend on the interval excluding zero.

What "converges" is doing here, and what it is not. The word appears elsewhere in this paper and carries more than the design delivers. A limit is not observed; a tail is. The measured statement is that the five-seed mean gap is below the 0.002 margin from about epoch 211 and that the tail is indistinguishable from zero at five seeds. A flat tail is also what a very slow decay looks like, and no test here separates the two: the stationarity result below cannot, because a ten-epoch window at five seeds cannot detect a change of a size that would still matter. The paper therefore reports a bounded residual, not a convergence rate, and does not quote +0.000231 as a floor.

Stationarity. A paired test of epochs 290–299 against 280–289 gives −0.00007 ($p = 0.555$; Wilcoxon 0.438; 1 of five seeds higher). The largest single step in the last ten epochs is 0.00046, or 4.3 per cent of the final-epoch seed standard deviation (`long_horizon_stationarity.csv`). The trajectory has stopped resolving above population noise, so 300 epochs suffices here and 100 does not. But "stops resolving" is a statement about the measurement, not about the process.

The curve fit, reported and not trusted. AIC and BIC prefer decay-to-constant over exponential decay decisively (ΔBIC = 24.1 from the peak), but the constant it estimates is *negative* with an interval excluding zero while the observed tail is positive: a single exponential cannot represent a series that first rises, so the constant absorbs the misfit rather than estimating an asymptote (`docs/STATS.md` §3.1; `long_horizon_decay_fits.csv`). The comparison answers which curve fits better, not what the asymptote is.

What this establishes. That the exposure gap at the shipped configuration is a cold-start property that disappears, and that a hundred-epoch figure describes a transient. Not that the mechanism is the one H3 offers, since the cell that was meant to test it turns out at this horizon to be the same process running more slowly (§5.6), nor that convergence holds elsewhere, since the three-hundred-epoch run exists only at the shipped cell.

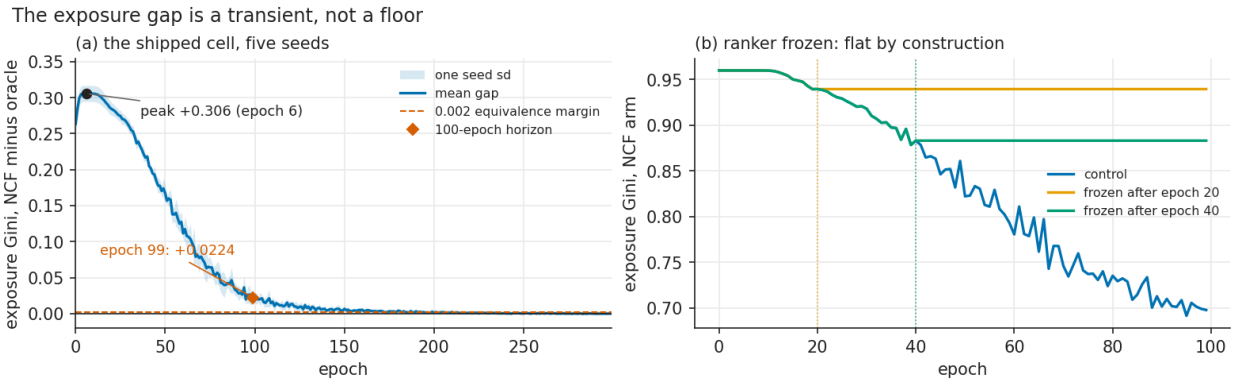


Figure 1: The paper's central result, which was previously a table of eight numbers. Panel (a): the five-seed mean exposure gap between the learned ranker and an oracle over 300 epochs, with a one-standard-deviation seed band, the peak at epoch 6 marked, the 0.002 equivalence margin as a dashed rule, and the shipped hundred-epoch point highlighted. The shape is the finding: a rise, a monotone decay over the range the data resolve, and a tail inside the margin. Panel (b): the same arm with the ranker's weights frozen after epochs 20 and 40, seed 42. Both frozen series are exactly flat, and that is a property of the design rather than a result: freezing the weights freezes the shortlists, so exposure cannot move once training stops. The panel is included because it makes the limitation of the intervention visible. Sources: `rigor/long_horizon_gap_series.csv`, `rigor/latefreeze/`.

5.4 The convergence claim as an equivalence test

"Below 0.001 and indistinguishable from flat" is not a statement with a defined meaning, so it is replaced by one. The margin was declared at 0.002 Gini before the test was run (`docs/PREREGISTRATION.md` E5), and the replicate is the seed, not the epoch: the fifty tail values are one trajectory, so their spread is not the sampling scale.

Two one-sided tests against ± 0.002 both reject. With a seed sd of 0.000340, a standard error of 0.000152 and four degrees of freedom, $t(\mu > -0.002) = 14.7$ and $t(\mu < +0.002) = 11.6$; the 90% interval over seeds is $[-0.000094, +0.000555]$, wholly inside the margin. The tail is *equivalent* to zero at a pre-declared margin, which is a stronger and more checkable statement than an interval that happens to include zero, and it is the one claim here that the tail supports: the 95% interval itself includes zero (§5.3).

The same test gives a better summary of a transient than any single-epoch value: the epoch at which the gap first stays below the margin for the remainder of the run, per seed, is 245, 262, 198, 223 and 211. That range, 198 to 262 epochs rather than the 120 to 200 the pre-registration estimated, is the quantity to quote when a reader asks how long the transient lasts.

5.5 The institutional grid was measured in the inert regime

One finding in this session has no pre-registration and belongs in the main text anyway, because it changes what H4 means. The institutional comparison in Appendix A was generated in October at `admissible_fraction` $= 0.2$. That is the *inert* gate: the configuration §4 describes as one where θ never truncates a shortlist and the market never responds to its own outcomes. The shipped value is now 0.02.

Both statements are verifiable from the repository. Re-running seeds 42–46 at 0.2 repro-

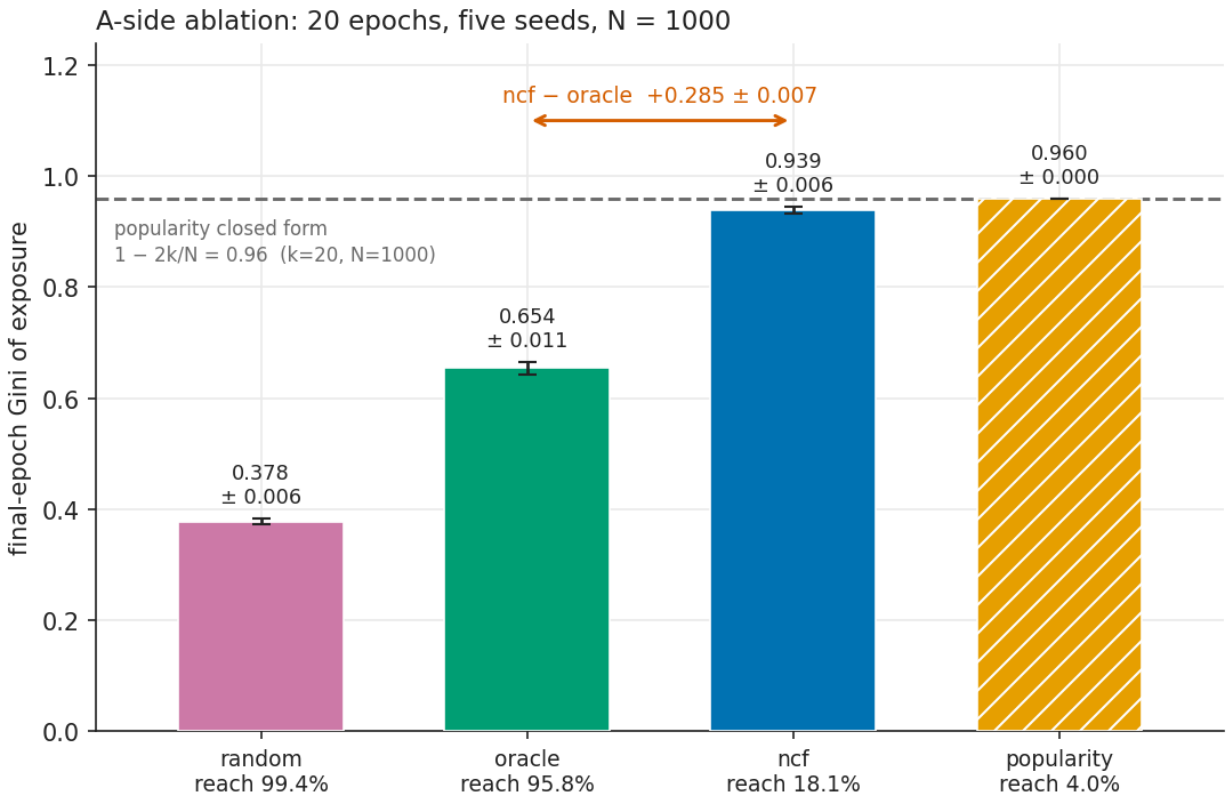


Figure 2: The four arms at the end of a 20-epoch ablation, five seeds. Earns its place as the only committed artifact showing the H3 inequality against both controls at once: a random baseline, which measures distance from uniformity rather than harm, and an oracle, which is unattainable. The popularity arm sits at the analytic closed form for a popularity ranker, checking that the exposure machinery behaves as designed; the reach labels are the mechanism's own diagnostic and are what §5.6 interrogates. Source: `data/processed/ablation/`.

duces all 200 rows of `data/processed/institutional/grid.csv` exactly, to the last digit, on `welfare_common_a`, `welfare_diff_residual` and `gini_a`. Re-running the same seeds at the shipped 0.02 does not: the optimum cell reads −17.23 against the committed +64.62. The regime, not the code and not the seeds, is the whole difference.

So the interior optimum H4 reports is a property of a market that does not learn from its own clearing. That much was established by reproducing the committed grid; the next question is whether the optimum survives at the gate the paper actually runs, and it does not.

The grid re-run at the shipped gate. The full ten-by-four grid was run at `admissible_fraction` = 0.02 on the same five seeds (`data/processed/rigor/h4_shipped_gate/`). **No cell of the forty beats Environment A.** At $c = 0.01$ the difference is close to monotone decreasing in subgraph size, from −0.23 at size 0.02 to −40.22 at 1.00, and the cell that was the optimum under the inert gate is number 7 of ten by mean, at −17.23. At $c = 0.05$, 0.10 and 0.25 the best cell is negative as well (−0.25, −0.28, −0.37).

This result was not pre-registered, and that matters for how much weight it carries: it is recorded in `docs/PREREGISTRATION.md` E6 with that label, on the reasoning that a first measurement of a quantity with no prior is weak evidence but not biased evidence, and that presenting it as a

Knob	Level	Gap	95% BCa	95% t	C1	C2	C3	C4
taste/sigma	0.2	+0.0268	[+0.0209, +0.0326]	[+0.0175, +0.0361]	yes	yes	yes	yes
	0.5	+0.0260	[+0.0164, +0.0348]	[+0.0110, +0.0409]	yes	yes	yes	yes
	1.0	+0.0265	[+0.0184, +0.0330]	[+0.0148, +0.0382]	yes	yes	yes	yes
shock/scale	0.5	+0.0097	[+0.0062, +0.0116]	[+0.0058, +0.0137]	yes	yes	**no**	yes
	1.5	+0.0260	[+0.0164, +0.0348]	[+0.0110, +0.0409]	yes	yes	yes	yes
	3.0	+0.0470	[+0.0192, +0.0665]	[+0.0109, +0.0831]	yes	yes	yes	yes
beta/alpha	1.0	+0.0340	[+0.0248, +0.0433]	[+0.0191, +0.0489]	yes	yes	yes	yes
	2.0	+0.0260	[+0.0164, +0.0348]	[+0.0110, +0.0409]	yes	yes	yes	yes
	3.0	+0.0224	[+0.0194, +0.0243]	[+0.0186, +0.0262]	yes	yes	yes	yes
delta	0.0	+0.0229	[+0.0157, +0.0291]	[+0.0122, +0.0337]	yes	yes	yes	yes
	0.5	+0.0260	[+0.0164, +0.0348]	[+0.0110, +0.0409]	yes	yes	yes	yes
	1.0	+0.0263	[+0.0236, +0.0301]	[+0.0212, +0.0314]	yes	yes	yes	yes

Table 3: The sensitivity grid: one knob at one level per row, the rest at shipped values, five seeds, 100 epochs. Gap is the paired exposure difference, with its 95% BCa bounds and its 95% Student-t bounds printed side by side, because the C1 verdict must not rest on the construction the paper elsewhere says undercovers at five seeds. The shipped level of each knob is italicised and appears in three axes with identical numbers, which checks that the axes are independent. C1–C4 are the claims of `docs/STATS.md` §4.1, bold where a claim fails; C1 holds on both constructions in all twelve cells.

prediction would be worse than the breach. It is also the only measurement of this quantity.

The reading is therefore plain, and it costs the paper one of its two results. H4's interior optimum is a property of the regime where the threshold gate never binds and the market never responds to its own outcomes. At the configuration the rest of the paper runs, Environment B loses everywhere and loses more as it grows, and the thin-market penalty dominates the information gain at every size tested. The paper does not search for a cost or a size at which B wins; that search is what the pre-registration discipline exists to prevent, and the honest report is a refutation in the shipped regime alongside a positive result in a regime the paper does not otherwise use.

5.6 Robustness across the parameter sweep

The hypotheses above were each measured at one setting of four modeling knobs. This subsection asks where they hold: twelve cells, one knob moving at a time, four claims fixed with their thresholds before the cells were read (Table 3).

Holds everywhere. The H3 ordering holds in all twelve cells, positive with an interval excluding zero, reversing nowhere (C1). It holds on both constructions: every cell's Student-t interval excludes zero as well as every cell's BCa interval, and the two are printed side by side (Table 3). The gap is small by the pre-fixed ceiling in all twelve (C2), though the largest cell's upper bound reaches +0.0665 and is the weakest case. Clearing is non-degenerate in all twelve (C4). Neither the ordering nor the existence of a clearing market is an artifact of the shipped corner.

What does not, at this horizon. One cell fails the reach claim at 100 epochs: at the lowest chemistry noise the ranker reaches 0.4616 of agents against 0.9566 shipped, both fractions of the population (C3). The same comparison in counts is 462 agents against 957, out of $N = 1000$; reach is reported as a fraction everywhere else in this paper and is a fraction here. The gap there is small and positive (+0.0097), and the reading the sweep supports is that the cell is at a different point

in the same process rather than at the end of a different one.

And what a longer horizon does to that reading. The cell was then run to 300 epochs (`data/processed/rigor/c4_shock0.5/`; five seeds, 1 thread). Reach climbs from 0.041 at epoch 0 to 0.269 at epoch 49, 0.457 at epoch 99, and 0.490 at epoch 299. The gap peaks at +0.0579 at epoch 10 and decays to −0.0012 at epoch 299, proportionally as complete a decay as the shipped cell's, from a peak a fifth as large.

So 0.4616 was reach at *epoch 99*, not a ceiling. The sweep therefore does not identify a cell where the mechanism is absent, only one where it runs more slowly. That withdraws the conclusion-reversing claim this paper made and leaves the mechanism untested rather than vindicated, since a cell where reach climbs slowly is not a cell where reach is held fixed. At 100 epochs the filter learns coverage in every cell, at rates spanning a factor of two, and none was run long enough to show where the convergence it drives stops.

How large. The gap ranges from +0.0097 to +0.0470, a factor of about 5, with no change in sign; the shipped cell sits mid-range. Quote the range with its condition, or the shipped cell with its regime, never +0.025 as "the" gap.

A documented condition that does not reproduce. The configuration states, and a guard test pins, that the hypergamy margin δ has power only when $\beta/\alpha \geq 2$. Re-measuring shortlist overlap at the shipped heterogeneity and noise reverses the ordering: $\beta/\alpha = 1$ gives δ *more* power than 3, and all three levels clear the threshold the guardrail treats as meaningful (`docs/STATS.md` §4.2; `sensitivity/delta_power.csv`). The guard test is not wrong; it was measured on a noise-free homogeneous population, and the rule of thumb describes that population rather than the one the experiments run on.

What the sweep does show is that δ moves no headline: the gap is +0.0229 at 0, +0.0260 at 0.5 and +0.0263 at 1.0, a spread of 0.0034 against BCa intervals about 0.013 wide, while the weight ratio moves it 3.5 times as far and chemistry noise 11 times as far. The defensible statement is that no claim here rests on δ at any weight ratio tested, not that the results require $\beta/\alpha \geq 2$.

Refuted? The sweep refutes the unconditional form of the cold-start explanation and the stated β/α condition. It does not refute H3's ordering, the one result that survives every cell.

6 Limitations

Ordered by how much each constrains what the results can mean, not by how easy it is to fix.

Every parameter is a modeling choice. Nothing is calibrated to data: the trait distributions are drawn for tractability, the taste and chemistry parameters have no empirical anchor, and neither the signalling cost, the defection penalty, nor the entry filter is estimated. The results bound what this mechanism *can* do under a stated utility and say nothing about whether any population sits at these parameters. Because traits are z-scored before entering a utility, only rank order matters, but a distribution with no variance would still destroy its term, so the defaults are deliberately not self-report-shaped and the attractiveness result is an upper bound on its influence rather than an estimate. The mitigation is that the two most consequential knobs are reported as a function of their value. It is partial: a sweep over four knobs is not a calibration.

Five seeds carry most headlines, twenty carry six. The power analysis says five is adequate for sign and inadequate for width on two quantities, and both matter. An interval on a model's seed-averaged quantity is not an interval on the model's answer. The five seeds were also not chosen for this purpose; they predate the twenty-seed run, which shows them to be an unusually low-variance block for two of the seven tracked quantities (`docs/STATS.md` §2.2). That finding is reported and is not repaired: the interval arithmetic every table uses was changed to the

construction with nominal small-sample coverage, but the seeds themselves were left alone, so any claim whose variance happens to be underestimated by this particular block still reads as tighter than it is. The honest reading is that the two affected quantities, algorithmic loss and clearing rate, carry no headline here, and that a future study should choose seeds for the purpose rather than inherit them.

The H4 optimum carries a sharper caution than a multiplicity correction can address. It was selected from forty cells on mean *and* low variance, and Bonferroni corrects the first without undoing the second: on fresh seeds the cell replicates its rank and loses most of its claimed precision, its seed sd rising from 9.86 to 46.42. And the whole grid was measured in a gate regime that does not respond to the market, which is not the configuration this paper otherwise runs. At the shipped gate no cell of the forty beats Environment A at all, so the interior optimum is a result about a regime rather than about the model (§5.5).

The committed 300-epoch run does not reproduce. Its NCF arm cannot be regenerated from this code at 1, 2, 4, 8, 16 or 24 threads, nor from the pre-optimization source at 24; its other three arms regenerate exactly. The divergence is at the seventh decimal at epoch 5 and grows through the feedback loop, which is the signature of a different reduction order rather than a different model (`docs/REVIEW_RESPONSE.md`, A3). The convergence numbers this paper quotes therefore come from a regeneration at one thread rather than from the committed run; the two agree within tolerances fixed before the regeneration was read (`docs/PREREGISTRATION.md` E7). One other archival directory is in a milder version of the same position: `rigor/base/`, which carries the twenty-seed headline, differs from a fresh run at the same thread count in the fourth decimal.

The thread count is part of the run's identity, and that was found late. Torch's intra-op reductions are not bitwise associative, so the same seed at one thread and at eight gives different numbers; across a hundred epochs of this feedback loop the spread across thread counts reaches 0.0070–0.0154 in exposure Gini, against a between-seed standard deviation of 0.0159 over twenty seeds. Every configuration now pins `ncf.num_threads` to 1 and the contract is tested, but the reproducibility guarantee this project claims is a guarantee at a named thread count, not an unconditional one (`docs/INTERFACES.md` §3; `data/processed/rigor/thread_sensitivity.csv`).

One population size carries the headline. $N = 1000$ is the only size at which the full comparison was run, because the full-information benchmark needs an $O(N^2)$ preference matrix and is capped at $N = 5000$. Clearing rate is not comparable across population sizes at fixed k: it falls roughly as $1/N$, and scaling k to compensate makes the swipe log grow superlinearly. The clearing figures are properties of $N = 1000$ at $k = 20$ and do not transport.

There is no closed loop through preferences. The ranker changes who is visible, not what anyone wants: the preference generator takes no shown-set argument. A reader looking for the classic feedback story will not find it here. A different loop exists, through the acceptance threshold, and it is live at the shipped configuration; but it relaxes within a few epochs, so even the loop that is present is a transient. This is the largest gap between this model and the recommender literature it engages, and it is a structural absence rather than a measurement problem.

There is no attrition. Nobody joins or leaves in response to outcomes, so there is no selection at the extensive margin and a market that deteriorated enough to empty out could not show it. This is why one hypothesis is untested (§A.5), and it bounds the welfare reading: a mechanism that repelled participants would look free.

The defection penalty prices defection without deterring it. It is charged after a match dissolves; no agent proposes differently, sets a different threshold, or stays longer because of it. Making it deter would mean feeding it into the proposal decision, which is a different model whose institutional result has not been measured.

The benchmark is not an achievable institution. The oracle ranks by true utility, so every gap measured against it is a distance from an unattainable bound rather than a choice an agent could make. Conversely, the institutional comparison scores B against an oracle A, and against an NCF-mediated A the ordering reverses. Both facts bound what the A/B result means: it is about the quality of A's visibility, not institutional form in general.

Truncation is imposed, not strategic. Agents receive their top-k list and do not game it, so the strategic theory of truncation does not apply directly, and the model cannot say what happens if agents report their lists instead.

Validation establishes verification, not validity. Agreement with closed-form results shows the machinery is correct; it establishes nothing about the recommender, because the deficit those theorems characterize is measured under a compatibility constraint on a single scalar, which is exactly what makes it tractable, while this model's constraint is visibility through an estimated multi-attribute utility. A matching deficit is not evidence that the recommender's unmatched mass has a closed form. It does not appear to have one.

Nothing here is a claim about people. The intent type has no instrument, the utility has no estimated weights, and the environments are model constructs. Neither is ethically superior: B carries an entry filter that excludes agents, and the model prices that exclusion rather than endorsing it.

7 Conclusion

A learned ranker between the two sides of a matching market concentrates exposure relative to an oracle, and the concentration is a start-up cost rather than a standing feature. That is the paper's finding, and it is a finding about *when* the measurement is taken as much as about what is measured: at the shipped hundred-epoch horizon the gap is $+0.0237$ over twenty seeds and looks like a floor; at three hundred epochs a pre-declared equivalence test at a 0.002 margin puts it inside the margin. The hundred-epoch figure is therefore far from where the series settles, though the ratio between the two is not itself a measured quantity. A literature that reports single-horizon comparisons cannot distinguish a transient from a bias, and this mechanism supplies a case where the distinction is decisive. What is not established is the limit: a flat tail is compatible with a very slow decay, and separating them needs a horizon this study did not run.

Three qualifications travel with it, and two of them are retractions. The ordering holds in every cell of a twelve-cell sweep, on both interval constructions, but its size varies fivefold, so no single number is "the" gap. The mechanism offered, that the filter learns whom it can represent, is not established here: the cell that was meant to isolate a market without coverage learning turns out at a longer horizon to be the same process running more slowly, and freezing the ranker cannot test the claim because freezing the weights freezes the shortlists. And the institutional companion does not survive where the paper runs: an interior optimum exists in the gate regime that ignores the market's own outcomes, and at the shipped gate no cell of forty beats the global market at any size or cost. A companion result that requires a configuration the rest of the paper does not use is a result about that configuration.

What would move this from a model result toward a claim about markets is calibration, and the paper does not attempt it. What would move the exposure result from a transient to a mechanism is an intervention that separates training from reach, which this model cannot supply. Until then the results bound what this mechanism can do under a stated utility, and the bound is the contribution.

A Appendix: hypotheses reported in full

This appendix holds the five hypothesis subsections whose verdicts are summarized in §5. Nothing here is new; it is the same analysis at the same length, moved so that the main text's argument is carried by the results that survive it.

A.1 H1: a cost on proposing selects on intent

Under test. Mean received intent rises with the signalling cost c, and low-intent proposers exit. Kill condition: the mean intent is flat in c.

Measured. The direction holds: mean received intent is monotone increasing over the swept grid (Figure 3), so the kill condition is not met. The figure also shows the real finding. The analytic threshold $\tau^*(c) = c/(\bar{V}p)$ is evaluated at a prior success probability, while the reciprocity the market delivers is a small fraction of that prior. A threshold evaluated at a probability the market does not supply cuts more aggressively than the realized one, so the measured points sit between the pooling line and the analytic curve rather than on it. Selection happens, later and more weakly than the closed form predicts.

Scope. The hypothesis is testable only where the market stays open. Carrying measured rather than prior reciprocity, the market closes for good a short way above the bottom of the grid (`docs/THESIS.md`, H1 caveat; `docs/THEORY.md` §10.2), so the surviving statement is narrower than the one posed.

Refuted? No. Flatness is required and there is none. What is refuted is the implied magnitude. No interval is registered for this quantity.

A.2 H2: threshold inflation as defence against pooling

Under test. The acceptance threshold θ rises with received volume *net of* the mean intent of those proposals, and stops rising when c makes the signal informative. Kill conditions: unrelated to volume, or still rising at high c.

Measured: there is no instrument. The shipped θ update reads one thing, the count of proposals received through clearing, and reads it only as a yield denominator. It never reads received intent. The count is bounded above by one per proposer while θ starts far above the unit interval a yield can occupy, so when the update fires it can only lower θ, for a minority of acceptors, by a fixed step. No path exists by which a net-of-intent slope could be positive, because the intent term is not in the code. The observed decline of θ in volume is not evidence either way: it reflects which acceptors the update reaches, and holding reach fixed the association vanishes.

Would it survive if the mechanism existed? Partly, and self-defeatingly. Lengthening the shortlist scales volume but also moves the gate, so it is not a clean instrument. A flooding term added by hand does produce the predicted sign, but only at the largest weight tested and the smaller shortlist, and at that weight it raises the most-swiped acceptors' thresholds until reciprocity falls to zero and the market closes (`data/processed/flooding/`; `docs/THESIS.md`, H2). A defence against pooling that empties the market is not a defence.

Refuted? Formally yes, substantively uninformative: the mechanism is absent from the model rather than present and ineffective. Reporting this as a refutation would be a category error and as support would be worse. H2 is untestable under the shipped rule, and that is the finding.

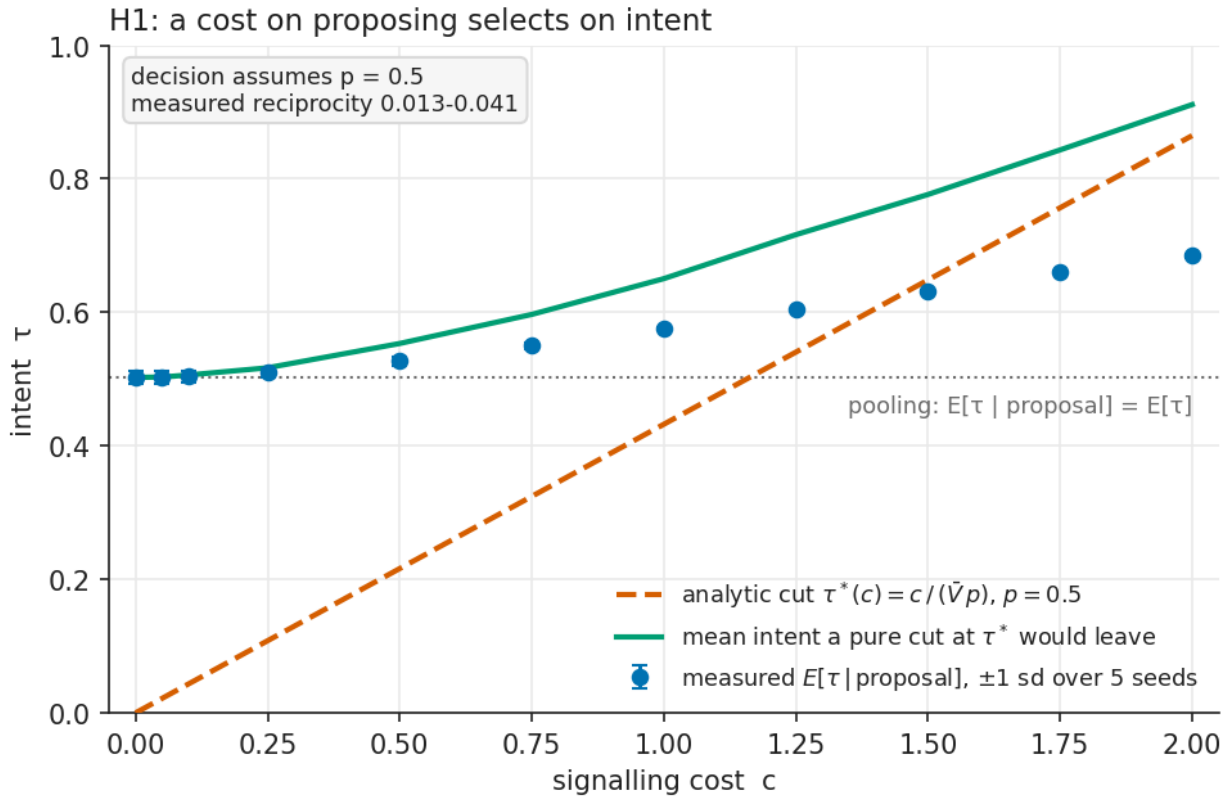


Figure 3: H1: measured mean intent of received proposals against signalling cost, with the analytic cut $\tau^*(c)$ and the intent a pure cut would produce. Earns its place as the only committed artifact for H1, and because it shows the discrepancy the text reports: the analytic curve assumes a reciprocity the market does not supply, so the measured points lie between the pooling line and the curve. No interval is registered for this quantity.

A.3 H4: an interior optimum in the institutional comparison

Under test. Welfare under Environment B exceeds welfare under A for subgraph sizes above a critical mass and falls below it for small sizes, at fixed cost. Kill condition: the ordering is monotone in size: B always wins, so the thin-market penalty does nothing, or always loses (the information gain does nothing). Welfare is scored on a common basis, with agents B excludes priced at their counterfactual A outcome.

Measured. Survives on one slice, killed on the others. At $c_B = 0.01$ against an oracle A, the optimum is size_fraction 0.65 with a residual welfare difference of $+64.62$, t interval $[+52.37, +76.86]$ (`docs/STATS.md` §1, row 7b), and the ordering there is non-monotone, which is the shape H4 asks for (Figure 4).

The selection correction the claim requires. The optimum is the maximum of forty cells, so an uncorrected interval on it is not an interval on anything. Corrected over that family, the Benjamini–Hochberg q-value is 3.88×10^{-4}, Holm rejects, and the Bonferroni t interval widens to $[+28.80, +100.43]$ (`docs/STATS.md` §5.3; `multiple_comparisons.csv`). At roughly fifteen standard errors ($64.62/(9.86/\sqrt{5}) = 14.65$) the correction is cheap.

The narrower finding. Of seven cells where B beats A, only two survive correction (0.65 and 0.50). The optimum is a two-cell feature, not a region; the curve beyond those cells is indistinguish-

able from zero. A caution the correction does not address: the optimum's standard deviation is the ninth smallest of the forty, at a little under a third of the median cell's spread, so it is a cell where five seeds happened to agree, which is why the t interval is the one to read.

What the correction does not fix. Bonferroni and Holm widen an interval for having searched a family; they do not undo the fact that the cell was selected on low variance as well as high mean. The interval above is therefore an honest width around a point that may be biased upward by the selection itself, and the two are not the same problem.

The re-estimate on seeds that took no part in the choice. That objection was answered in this session rather than deferred. The ten `size_fraction` levels at $c = 0.01$ were re-run on seeds 100–119, at the same gate as the original, after the prediction was committed (`docs/PREREGISTRATION.md` E2; `data/processed/rigor/c2_fresh_inert/`). Two things happened, and they point opposite ways.

The *location* replicated. On the fresh grid the cell at 0.65 still ranks first of ten, with positive values from 0.35 through 1.00, a maximum at 0.65, and negatives below 0.25: the non-monotone shape H4 asks for, on twenty populations that took no part in choosing the cell. Its interval there is $[+32.72, +76.17]$, excluding zero ($t = 5.25$ on 19 df).

The *magnitude and its precision* did not. The fresh estimate is +54.45 against +64.62, and, the larger point, the fresh seed sd is 46.42 against the original 9.86. The published interval was built on a standard deviation 4.7 times too small, because the cell was selected partly for having an unusually low one. That is the same pathology `docs/STATS.md` §2.2 found for algorithmic loss and clearing, appearing in the cell this hypothesis is built on.

So the correct reading is: an interior optimum at subgraph size 0.65 that replicates on fresh seeds, with a gain of roughly +54 rather than +65, and an uncertainty the published t interval understates by a factor of about five. A reader who wants one number should take the fresh one.

And it is an inert-regime result, which the shipped regime refutes. The whole grid above was measured at `admissible_fraction` 0.2, the configuration §5.5 shows to be deaf to the market's own outcomes; the shipped value is 0.02. That experiment was then run: the same ten-by-four grid at 0.02 on the same five seeds has **no cell of the forty beating Environment A**, at any size and any of the four costs, and at $c = 0.01$ the difference falls almost monotonically with size from -0.23 to -40.22. The cell that is the optimum here reads -17.23 there.

So H4 as stated is refuted at the configuration the rest of the paper runs, and what survives is a narrower statement: in a market whose threshold gate never binds, an interior optimum exists at a subgraph size around 0.65. A companion result that requires a configuration the main study does not use is a result about that configuration, and it should be read that way.

Killed elsewhere. At $c_B \geq 0.05$ B always loses: no size beats A beyond seed noise (`data/processed/institutional/`). Against an NCF-mediated A rather than an oracle A the ordering reverses and B always wins, so the result is about the quality of A's visibility, not subgraph size as such.

Refuted? Yes, on the $c_B \geq 0.05$ slice, and yes in that the hypothesis is a property of one cost level rather than of the model. What survives: at a cost low enough that entry screens little, an interior optimum exists and is two cells wide.

A.4 H5: the decomposition's interaction term

Under test. The welfare change from an exogenous status shift and from algorithmic mediation decomposes into additive and interaction parts, with the interaction estimated rather than assumed. The hypothesis does not predict its sign; it predicts the term is estimable and may be near zero.

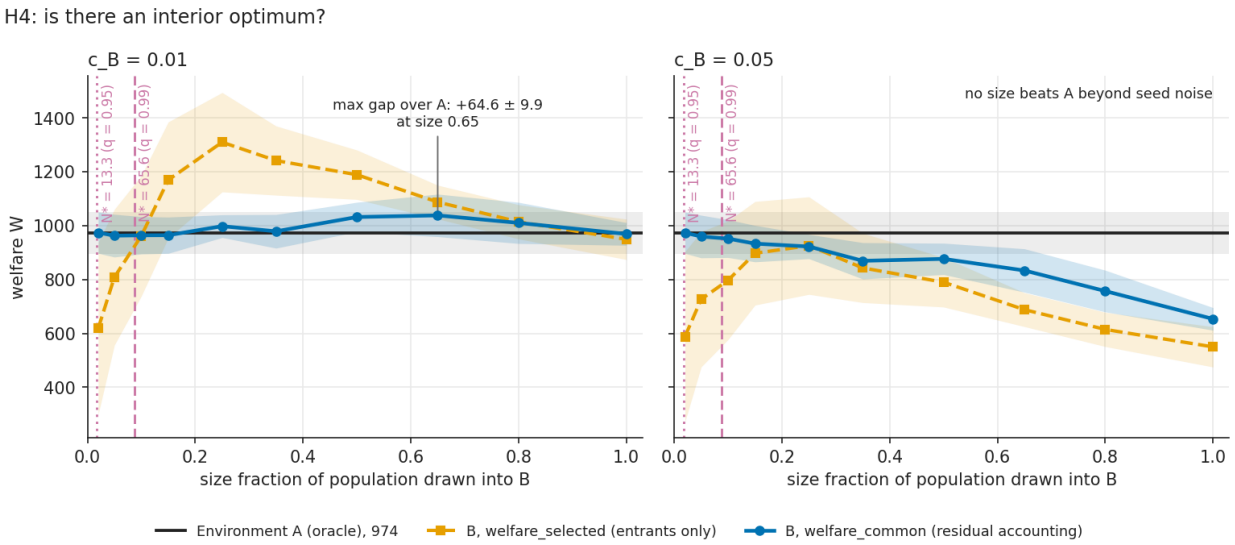


Figure 4: H4: welfare against the share of the population drawn into Environment B, at two signalling costs, with Environment A as a horizontal line. Earns its place because one frame carries all three elements of the verdict: the non-monotone shape at the low cost, the monotone collapse at the high cost where no size beats A, and the gap between selected and common accounting (dashed against solid), which is the selection artifact the comparability rule exists to price. Vertical rules mark analytic critical masses; they serve the thin-market penalty and do not explain the right-hand slope, which `data/processed/THIN_MARKET_README.md` attributes to top-k truncation.

Measured. The interaction is +16.70 with a t interval of $[-33.42, +66.82]$ and a BCa interval of $[-13.30, +53.05]$; its BH q-value over the three terms is 0.407 and it is not rejected (`docs/STATS.md` §5.4). Indistinguishable from zero, as H5 allowed.

The terms that are not zero. The algorithmic term is -717.61 $[-795.77, -651.37]$ and dominates: mediation costs an order of magnitude more welfare than the demographic shift moves it. The exogenous term is -21.10, and it is where the two methods disagree: its BCa interval $[-41.13, -3.42]$ excludes zero while its t interval $[-52.53, +10.32]$ and q-value of 0.204 do not. At five seeds with this skew the BCa interval is optimistic, so the defensible reading is a negative sign with unresolved magnitude. Figure 5 shows the arithmetic and the scale difference.

Refuted? No. The kill condition was incomparability of the factorial cells, which would leave the decomposition undefined; comparability is asserted per seed and holds. The measurement establishes a negative result reported honestly: drift and mediation act through separate channels here, and the claim that the algorithm amplifies the shift has no support.

A.5 H6: differential attrition by intent

Under test. High-intent agents leave Environment A at a higher rate than low-intent agents. Kill condition: attrition flat in τ, or the sign reversed.

Measured: not at all. The model has no attrition mechanism. Agents do not exit in response to outcomes; the only population change is generational turnover, which is not a decision and does not read τ. No committed artifact exists for this hypothesis and no quantity is registered, so no number is quoted.

What can be said without one. The sign of any measured correlation is determined by the exit

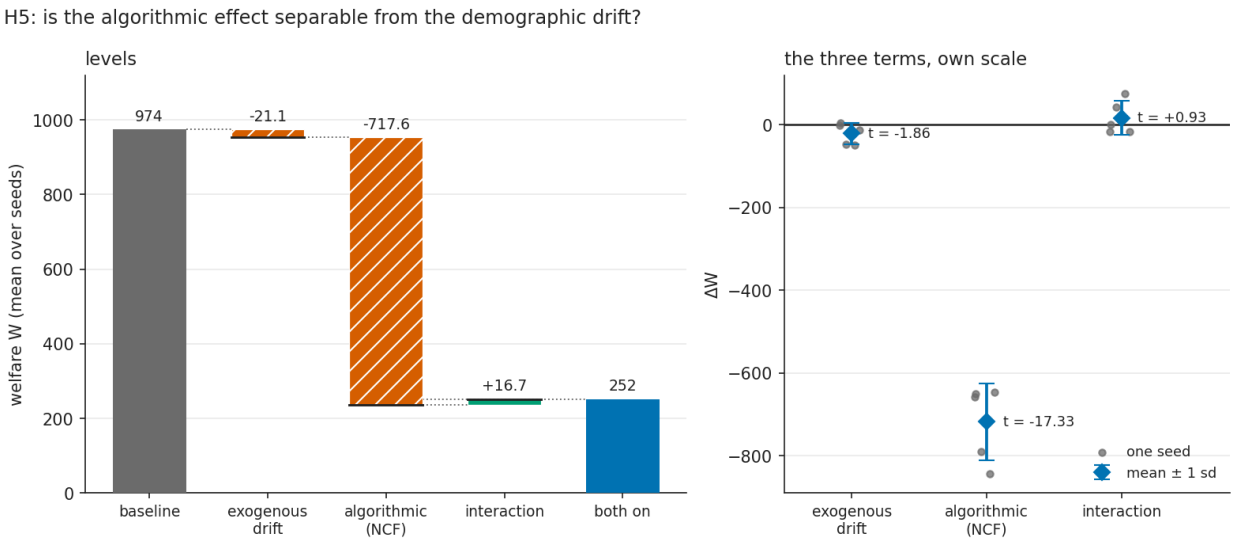


Figure 5: H5: the four factorial levels and the three decomposed terms. Earns its place because the left panel shows the decomposition's arithmetic, with the interaction as the residual closing the gap between the marginal effects and the joint effect, while the right panel puts the three terms on their own scale with per-seed points, which is what makes the scale difference between the algorithmic term and the other two visible. Without the left panel the terms cannot be checked to add up; without the right panel the interaction looks larger than it is. Source: `data/processed/institutional/factorial.csv`.

rule imposed, and the two examined point opposite ways: keying exit on being unmatched yields no association, since matching does not depend on τ; keying it on net payoff yields a strong positive one, but by cost burn rather than rejection: high-intent agents propose far more often and are matched at the same rate (`docs/THESIS.md`, H6). Neither tests the hypothesis, which is about a market with an exit margin.

Refuted? Neither confirmed nor refuted: untested, and reported as such. The cost-burn channel it exposes is the same one that kills H4 at high signalling cost.

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